Advanced Econometrics Lecture 14

dr Marcin Chlebus, dr Rafał Woźniak

Faculty of Economic Sciences, University of Warsaw

Warsaw

What if I told you that everything you knew was a lie?

A variable is *endogenous* if it is correlated with the disturbance. In the model

$$y = \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_K x_K + \varepsilon \qquad (1)$$

x_j is endogenous if $Cov[x_j, \varepsilon] \neq 0$.
x_j is exogenous if $Cov[x_j, \varepsilon] = 0$.

The OLS estimator will be consistent only if $Cov[x_j, \varepsilon] = 0$ for $j = 1, 2, \ldots, K$.

Cases in Which x and e are Correlated

There are several common situations in which the least squares estimator fails due to the presence of correlation between an explanatory variable and the error term. When an explanatory variable and the error term are correlated, the explanatory variable is said to be **endogenous**.

1. Measurement Error
2. Omitted Variables
3. Simultaneous Equation Bias
4. Lagged Dependent Variable Model with Serial Correlation

Source: Principles of Econometrics, 4th Ed., p. 405

Omitted variables in OLS

Problem and its consequences

- Omitted variables are insignificant
- Omitted variables are significant and orthogonal to dependent variables
- Omitted variables are significant and non-orthogonal to dependent variables

Omitted variables case – Principles..., 3rd Ed., p. 275

When an omitted variable is correlated with an included explanatory variable, then the regression error will be correlated with the explanatory variable, making it endogenous. The classic example is

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + e_i \tag{2}$$

with $EDUC =$ years of education. **What else affects wages? What else is omitted?**

This introspective experiment should be carried out each time a regression model is formulated.

We have omitted many factors: experience, ability, and motivation.
If we omit these factors, where do they go?

Like all omitted factors, they go into the error term e_i.
Do we expect $Cov[EDUC_i, \varepsilon i] = 0$?

Simultaneous Equation Bias – Principles..., 4th Ed., p. 406

Another situation in which an explanatory variable is correlated with the regression error term arises in simultaneous equations models. Students of economics deal with such models from their earliest introduction to supply and demand. Recall that in a competitive market, the prices and quantities of goods are determined jointly by the forces of supply and demand. Thus if P = equilibrium price and Q = equilibrium quantity, we can say that P and Q are endogenous, because they are jointly determined within a simultaneous system of two equations, one equation for the supply curve and the other equation for the demand curve. Suppose that we write down the relation

$$Q = \beta_1 + \beta_2 P + e \tag{3}$$

We know that changes in price affect the quantities supplied and demanded. But it is also true that changes in quantities supplied and demanded lead to changes in prices. There is a feedback relationship between P and Q. Because of this feedback, which results because price and quantity are jointly, or simultaneously, determined.

Instrumental variables in the simple linear regression model

Problems for least squares arise when x is random and correlated with the random disturbance e, so that $\mathbb{E}[xe] \neq 0$. Suppose, however, that there is another variable z such that

1. z does not have a direct effect on y, and thus it does not belong on the right-hand side of the model as an explanatory variable.
2. z is not correlated with the regression error term e. It is exogenous.
3. z is strongly (or at least not weakly) correlated with x, the endogenous explanatory variable.

A variable z with these properties is called an **instrumental variable**. This terminology arises because although z does not have a direct effect on y, having it will allow us to estimate the relationship between x and y. It is a tool, or instrument, that we are using to achieve our objective.

Example 1 – Example 16.5 from Mycielski

Usually, in economic datasets there is no information on ability. Ability affects either education attainment or wage. If instead of

$$\ln(\mathit{WAGE}) = \beta_0 + \beta_1 \mathit{EDUC} + \beta_2 \mathit{SEX} + \beta_4 \mathit{AGE} + \beta_5 \mathit{AGE}^2 + \beta_5 \mathit{ABILITY} + \varepsilon, \quad (4)$$

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$$\ln(WAGE) = \beta_0 + \beta_1 EDUC + \beta_2 SEX + \beta_4 AGE + \beta_5 AGE^2 + \beta_5 ABILITY + \varepsilon, \quad (4)$$

we would estimate

$$\ln(WAGE) = \beta_0 + \beta_1 EDUC + \beta_2 SEX + \beta_4 AGE + \beta_5 AGE^2 + u, \quad (5)$$

then we omit significant variable *ABILITY*, and the error in equation (5) is $u = \beta_5 ABILITY + \varepsilon$.

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Instrumental variables

- What should we use as an instrument?

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Instrumental variables

- What should we use as an instrument?
- parents' education level?

Example 2

Example 15.3 [Wooldridge, Introductory Econometrics, p. 494]

- Let's consider birth weight of newborns in regard of number of cigarettes packs smoked a day by their mothers:

$$log(bweigth) = \beta_0 + \beta_1 packs + u \tag{6}$$

- Variable *packs* might be correlated with unobserved variable for prenatal care.
- Instrument: price of a pack of cigarettes?

Example 3

We have state data from the 1980 census on the median dollar value of owner-occupied housing (*hsngval*) and the median monthly gross rent (rent). We want to model rent as a function of *hsngval* and the percentage of the population living in urban areas (*pcturban*):

$$rent_i = \beta_0 + \beta_1 housevalue_i + \beta_2 pcurban_i + \varepsilon_i, \tag{7}$$

where i indexes states and u_i is an error term.

- Because random shocks that affect rental rates in a state probably also affect housing values, we treat *hsngval* as endogenous.
- We believe that the correlation between *hsngval* and u is not equal to zero.
- On the other hand, we have no reason to believe that the correlation between *pcturban* and u is nonzero, so we assume that *pcturban* is exogenous
- In our dataset, we have a variable for family income (*faminc*) and for region of the country (*region*) that we believe are correlated with *hsngval* but not the error term.

Source: Stata Help (ivregress)

Example 4

Acemoglu, Daron, Simon Johnson, and James Robinson. 2005. "The Rise of Europe: Atlantic Trade, Institutional Change, and Economic Growth." American Economic Review, 95 (3): 546-579. (link)

We can test the idea that West European growth after 1500 was due primarily to growth in countries involved in Atlantic trade or with a high potential for Atlantic trade.

An important concern with the results reported so far is endogeneity. Being an Atlantic trader is an ex post outcome, and perhaps only countries with high growth potential - or those that were going to grow anyway - engaged in substantial Atlantic trade and colonial activity.

We use a geographic measure of potential access to the Atlantic, Atlantic coastline-to-area ratio, which gives positive Atlantic trade potential to all these countries.
Alternatively, we could use the Atlantic coastline-to-area measure as an instrument for the Atlantic trader dummy.

Example 5

Steven D. Levitt, *The Effect of Prison Population Size on Crime Rates: Evidence from Prison Overcrowding Litigation*, The Quarterly Journal of Economics, Volume 111, Issue 2, 1 May 1996, Pages 319–351.

Simultaneity between prisoner populations and crime rates makes it difficult to isolate the causal effect of changes in prison populations on crime. To break that simultaneity, this paper uses prison overcrowding litigation in a state as an instrument for changes in the prison population.

Example 6

Steven D. Levitt, *Using Electoral Cycles in Police Hiring to Estimate the Effect of Police on Crime*, The American Economic Review, Vol. 87, No. 3 (Jun., 1997), pp. 270-290.

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- Detroit has twice as many police officers per capita as Omaha, and a violent crime rate over four times as high, but it would be a mistake to attribute the differences in crime rates to the presence of the police.

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- Detroit has twice as many police officers per capita as Omaha, and a violent crime rate over four times as high, but it would be a mistake to attribute the differences in crime rates to the presence of the police.
- Similarly, within a particular city, if more police are hired when crime is increasing, a positive correlation between police and crime can emerge, even if police reduce crime.

Example 6 contd.

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- Instrumental variables?

Example 6 contd.

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- Instrumental variables?
- The instrument employed in this paper is the timing of mayoral and gubernatorial elections.

Examples of clever instruments

https://www.quora.com/What-are-some-examples-of-really-clever-instrumental-variables-approaches-in-econometrics
(link)

IV estimation in a general model

$$y = \beta_1 + \beta_2 x_2 + \ldots + \beta_G x_G + \beta_{G+1} x_{G+1} + \ldots + \beta_K x_K + e$$

- constant, $x_2, \ldots, x_G$ are exogenous
- $x_{G+1}, x_{G+2}, \ldots, x_K$ are endogenous, i.e. $K - G$ endogenous vars.

1. We need $L \geq K - G$ instrumental variables
2. First step of the 2SLS $\forall j = 1, 2, \ldots, K - G$

$$\begin{aligned} x_{G+j} &= \gamma_{1j} + \gamma_{2j} x_2 + \ldots + \gamma_{Gj} x_G + \theta_{1j} z_1 + \ldots + \theta_{Lj} z_L + v_j \\ \hat{x}_{G+j} &= \hat{\gamma}_{1j} + \hat{\gamma}_{2j} x_2 + \ldots + \hat{\gamma}_{Gj} x_G + \hat{\theta}_{1j} z_1 + \ldots + \hat{\theta}_{Lj} z_L + v_j \end{aligned}$$

3. Second step of the 2SLS

$$y = \beta_1 + \beta_2 x_2 + \ldots + \beta_G x_G + \beta_{G+1} \hat{x}_{G+1} + \ldots + \beta_K \hat{x}_K +$$

Source: Principles of Econometrics, 3rd ed., p. 285

An illustration using simulated data ([2], p. 280)

Let's artificially create a sample of $N = 100$ observation from the true model

$$y = 1 + x + e, \tag{8}$$

where x and e are normally distributed variables with true means 0 and variances 1. We set the population correlation between x and e to be $\rho_{xe} = 0.6$. Least squares estimation yields

$$\underset{(se)}{\hat{y}_{OLS}} = \underset{(0.088)}{0.9789} + \underset{(0.090)}{1.7034}x \tag{9}$$

The estimated slope is far from the true value $\beta_2 = 1$.

An illustration using simulated data – contd. ([2], p. 280)

Let's generate three instrumental variables

1. $z_1: \quad Corr(x, z_1) = 0.5, \quad Corr(z_1, e) = 0$
2. $z_2: \quad Corr(x, z_2) = 0.3, \quad Corr(z_2, e) = 0$
3. $z_3: \quad Corr(x, z_3) = 0.5, \quad Corr(z_3, e) = 0.3$

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$$\underset{(se)}{\hat{y}_{IV\_z_1}} = \underset{(0.109)}{1.1011} + \underset{(0.195)}{1.1924}x \tag{10}$$

$$\underset{(se)}{\hat{y}_{IV\_z_2}} = \underset{(0.256)}{1.3451} + \underset{(0.797)}{0.1724}x \tag{11}$$

$$\underset{(se)}{\hat{y}_{IV\_z_3}} = \underset{(0.095)}{0.9640} + \underset{(0.172)}{1.7657}x \tag{12}$$

Comment on the s.e. and the parameters' estimates

The importance of using strong instruments ([2] p. 279)

These new estimators have the following properties:

- They are consistent, if z is exogenous
- In large samples the instrumental variable estimators have approximate normal distributions. In the simple regression model

$$\hat{\beta}_2 \sim N\left(\beta_2, \frac{\sigma^2}{r_{zx}^2 \sum (x_i - \bar{x})^2}\right) \tag{13}$$

  where r_{zx}^2 is the squared sample correlation between the instrument z and the random regressor x.
- The error variance is estimated using the estimator

$$\hat{\sigma}_{IV}^2 = \frac{\sum (y_i - \hat{\beta}_1 - \hat{\beta}_2 x_i)^2}{N - 2} \tag{14}$$

Why shouldn't we use the 2-step procedure singlehandedly?

- The point of using the 2SLS estimator is the consistent estimation of $\hat{\beta}_{2SLS}$ in a model containing response variable y and regressors X, some of which are correlated with the disturbance process u.
- The prediction of that model involve the original regressors X, not the instruments $\hat{X}$.
- Although from a pedagogical standpoint we speak of 2SLS as a sequence of first-stage and second-stage regressions, we should never perform those two steps by hand.
- If we did so, we would generate predicted values $\hat{X}$ from the first-stage regressions of endogenous regressors on instruments and then run the second-stage OLS regression using those predicted values.
- Why should we avoid this? Because the second stage will yield the incorrect residuals

$$\hat{u} = y - \hat{X}\hat{\beta}_{2SLS} \quad \text{rather than} \quad \hat{u} = y - X\hat{\beta}_{2SLS} \tag{15}$$

Goodness-of-fit with IV estimates ([2] p.286)

We discourage the use of measures like R^2 outside the context of least squares estimation. When there are endogenous variables on the right-hand side of a regression equation, the concept of measuring how well the variation in y is explained by the x variables breaks down, these models exhibit feedback.

This logical problem is paired with a numerical one. If our model is

$$y = \beta_1 + \beta_2 x + e, \tag{16}$$

then the IV residuals are

$$\hat{e} = y - \beta_1 - \beta_2 x. \tag{17}$$

Many software packages will report the goodness-of-fit measure

$$R^2 = 1 - \frac{\sum \hat{e}_i^2}{\sum (y - \bar{y})^2}. \tag{18}$$

Unfortunately, this quantity can be negative when based on IV estimates.

Specification tests – [2] p. 286-287

- We have shown that if an explanatory variable is correlated with the regression error term, the least squares estimator fails.
- If a strong instrumental variable is available, the IV estimator is consistent and approximately normally distributed in large samples.
- But if we use a weak instrument, or an instrument that is invalid in the sense that it is not uncorrelated with the regression error, then IV estimation can be as bad as, or worse than, using the least squares estimator.
- Two other important questions that must be answered in each situation in which instrumental variables estimation is considered:

1. Can we test for whether x is correlated with the error term? This might give us a guide for when to use least squares and when to use IV estimators.
2. Can we test if our instrument is valid, and uncorrelated with the regression error, as required?

The Hausman test for endogenity – [2] p. 287

The null hypothesis is $H_0: \ cov(x,e) = 0$ against the alternative that $H_0: \ cov(x,e) \neq 0$. The idea of the test is to compare the performance of the least squares estimator to an instrumental variables estimator. Under the null and alternative hypotheses, we know the following:

- If the null hypothesis is true, both the least squares estimator b and the instrumental variables estimator β are consistent. Thus, in large samples the difference between them converges to zero. That is, $q = (b - \beta) \to 0$. Naturally, if the null hypothesis is true, use the more efficient estimator, which is the least squares estimator.
- If the null hypothesis is false, the least squares estimator is not consistent, and the instrumental variables estimator is consistent. Consequently, the difference between them does not converge to zero in large samples. That is, $q = (b - \beta) \not\to 0$. If the null hypothesis is not true, use the instrumental variables estimator, which is consistent.

Hausman test – [2] p. 287-288

An alternative form of the test is very easy to implement, and is the one we recommend. In the regression $y = \beta_1 + \beta_2 x + e$, we wish to know whether x is correlated with e. Let z_1 and z_2 be instrumental variables for x.

1. Estimate the first-stage model $x = \gamma_1 + \theta_1 z_1 + \theta_2 z_2 + v$ by least squares, including on the right-hand side all instrumental variables and all exogenous variables not suspected to be endogenous, and obtain the residuals

$$\hat{v} = x - \gamma_1 - \theta_1 z_1 - \theta_2 z_2 \tag{19}$$

   If more than one explanatory variable is being tested for endogeneity, repeat this estimation for each one.

2. Include the residuals computed in step 1 as an explanatory variable in the original regression, $y = \beta_1 + \beta_2 x + \delta \hat{v} + e$. Estimate this "artificial regression" by least squares, and employ the usual t-test for the hypothesis of significance:

$$H_0: \ \delta = 0 \quad \text{(no correlation between } x \text{ and } e\text{)} \tag{20}$$

$$H_1: \ \delta \neq 0 \quad \text{(correlation between } x \text{ and } e\text{)} \tag{21}$$

3. If more than one variable is being tested for endogeneity, the test will be an F-test of joint significance of the coefficients on the included residuals.

Testing for weak instruments – [2] p. 288

If the instruments we choose are weak, the instrumental variables estimator can suffer large biases and standard errors, and its large sample distribution might not be normal. Consider the model

$$y = \beta_1 + \beta_2 x_2 + \ldots + \beta_G x_G + \beta_{G+1} x_{G+1} + e \tag{22}$$

Suppose that $x_2, \ldots, x_G$ are exogenous and uncorrelated with the error term e, while x_{G+1} is endogenous. Further, suppose that we have one instrumental variable z_1. How can we determine whether it is a "strong" or "weak" instrument?

$$x_{G+1} = \gamma_1 + \gamma_2 x_2 + \ldots + \gamma_G x_G + \gamma_{G+1} x_{G+1} + \theta_1 z_1 + v \tag{23}$$

In order for z_1 to qualify as an instrument, the coefficient θ_1 in this reduced form equation must not be zero. The hypothesis must be soundly rejected. A common rule of thumb is that if the F-test statistic takes a value less than 10, or if the t-statistic is less that 3.3, the instrument is weak.

Sargan's test – testing instrument validity

A valid instrument z must be uncorrelated with the regression error term, so that $cov(z, e) \neq 0$. If this condition fails then the resulting moment condition, is invalid and the IV estimator will not be consistent. In order to compute the IV estimator for an equation with B possibly endogenous variables, we must have at least B instruments.

1. Compute the IV estimates β_K using all available instruments, including the G variables $x_1, x_2, \ldots, x_G$ that are presumed to be exogenous, and the L instruments $z_1, \ldots, z_L$.
2. Obtain the residuals $\hat{e} = y - \hat{\beta}_1 - \hat{\beta}_2 x_2 - \ldots - \hat{\beta}_K x_K$.
3. Regress $\hat{e}$ on all the available instruments described in step one.
4. Compute NR^2 from this regression, where N is the sample size and R^2 is the usual goodness-of-fit measure.
5. If all of the surplus moment conditions are valid, then $NR^2 \to \chi^2_{(L-B)}$. If the value of the test statistic exceeds the $100(1-\alpha)$th percentile (i.e., the critical value) from the $\chi^2_{(L-B)}$ distribution, then we conclude that at least one of the surplus moment conditions is not valid.

Method of Moments in the Simple Linear Regression

The definition of a 'moment" can be extended to more general situations. In the linear regression model $y = \beta_1 + \beta_2 x + e$, we usually assume that

$$\mathbb{E}[e_i] = 0 \Rightarrow \mathbb{E}[(y_i - \beta_1 - \beta_2 x_i)] = 0 \tag{24}$$

Furthermore, if x is fixed, or random but not correlated with e, then

$$\mathbb{E}[x_i e_i] = 0 \Rightarrow \mathbb{E}[x_i(y_i - \beta_1 - \beta_2 x_i)] = 0 \tag{25}$$

Equations (24) and (25) are moment conditions. If we replace the two population moments by the corresponding sample moments, we have two equations in two unknowns, which define the method of moments estimators for β_1 and β_2.

$$\frac{1}{N}\sum(y_i - \beta_1 - \beta_2 x_i) = 0 \tag{26}$$

$$\frac{1}{N}\sum x_i(y_i - \beta_1 - \beta_2 x_i) = 0 \tag{27}$$

Thus, under "nice" assumptions, the method of moments principle of estimation leads us to the same estimators for the simple linear regression model as the least squares principle.

Instrumental Variables and Method of Moments

If such a variable z exists, then we can use it to form the moment condition

$$\mathbb{E}[ze] = 0 \Rightarrow \mathbb{E}[z_i(y_i - \beta_1 - \beta_2 x_i)] = 0 \tag{28}$$

$$\frac{1}{N}\sum(y_i - \beta_1 - \beta_2 x_i) = 0 \tag{29}$$

$$\frac{1}{N}\sum z_i(y_i - \beta_1 - \beta_2 x_i) = 0 \tag{30}$$

Solving these equations leads us to method of moments estimators, which are usually called the instrumental variable (IV) estimators.

Exercise 10.7 (Hill, Griffiths & Lim, Principles of Econometrics)

A consulting firm run by Mr. John Chardonnay is investigating the relative efficiency of wine production at 75 California wineries. John sets up the production function

$$Q_i = \beta_1 + \beta_2 MGT_i + \beta_3 CAP_i + \beta_4 LAB_i + \epsilon$$

where Q is an index of wine output for a winery, taking into account both quantity and quality, MGT is a variable that reflects the efficiency of management, CAP is an index of capital input, and LAB is an index of labour input.

Because he cannot get data on management efficiency, John collects observations on the number of years of experience (XPER) of each winery manager and uses that variable in place of MGT. The 75 observations are stored in the file chard.dat.

Write the GMM conditions down

Literature

1. Martin, Vance, Hurn, Stan, & Harris, David (2013) *Econometric Modelling with Time Series : Specification, Estimation and Testing. Themes in Modern Econometrics.* Cambridge University Press, New York.
2. Principles of Econometrics, Third Edition, by R. Carter Hill, William E. Griffiths and Guay C. Lim
3. Mycielski J., *Skrypt do Ekonometrii*, in Polish
4. Christopher F. Baum, *An Introduction to Modern Econometrics Using Stata*, Stata Press, 2006